

Project Development Rules & Coding Standards

This document defines the standard development rules, coding conventions, file structures, naming rules, and project management guidelines for all software projects.

1. Variable Naming Rules

Rule
Use meaningful and readable variable names.
Use camelCase for variables and functions.
Boolean variables should start with is, has, can, or should.
Avoid single-letter variable names except for loops.
Constants must use UPPER_SNAKE_CASE.

2. File Naming Rules

Rule
Use lowercase file names with hyphen or underscore.
Avoid spaces in file names.
Each file should contain one primary responsibility.
Component files should match component names.
Keep folder structure clean and organized.

3. Class Naming Rules

Rule
Use PascalCase for class names.
Class names should describe business logic clearly.
One class should handle one responsibility.
Avoid overly large classes.
Use interfaces and abstract classes when necessary.

4. Database Development Rules

Rule
Use proper indexing for performance.
Never store plain text passwords.
Use migrations for schema updates.
Use snake_case for table and column names.
Always validate and sanitize user input.

5. API Development Rules

Rule
Use RESTful API structure.
Return consistent JSON responses.
Use proper HTTP status codes.
Implement authentication and authorization.
Document APIs using Swagger or Postman.

6. Git & Version Control Rules

Rule
Use meaningful commit messages.
Create feature branches for new development.
Never commit sensitive information.
Review pull requests before merging.
Keep the main branch stable.

7. UI/UX Development Rules

Rule
Ensure responsive design for all devices.
Maintain consistent color and typography.
Use accessible UI components.
Optimize loading performance.
Provide clear user feedback and validation.

8. Security Rules

Rule
Use environment variables for secrets.
Validate all API requests.
Implement role-based access control.
Use HTTPS in production.
Regularly update dependencies.

9. Project Management Rules

Rule

Define clear project scope and milestones.
Use task tracking tools.
Maintain proper documentation.
Conduct regular code reviews.
Test before deployment.